

• Homework 6

The Soyuz capsule orbits the Earth at an altitude of 400 km (in circular, equatorial orbit). Its main data are

$$m = 2400 \text{ Kg} \quad S = 3.8 \text{ m}^2 \text{ (aerodynamic surface)}$$

$$C_D = 1.341 \text{ (average value of } C_D \text{ along the entire flight)}$$

$$\tilde{L} = 0 \text{ (no lift)}$$

$$R_c = 2.235 \text{ m} \quad \left\{ \begin{array}{l} \text{radius of curvature at stagnation point} \\ \text{i.e. at the nose} \end{array} \right.$$

De-orbit follows a tangential braking Δv . Two cases are considered: (1) $\Delta v = 0.15 \frac{\text{km}}{\text{sec}}$ (2) $\Delta v = 0.3 \frac{\text{km}}{\text{sec}}$

For each of these two cases

- Find $v_{I,ini}$ and $\gamma_{I,ini}$ at $h_{E1} = 120 \text{ km}$, i.e. the inertial velocity magnitude and the related flight path angle at the entry interface
- Find $v_{R,ini}$ and $\gamma_{R,ini}$ associated with $v_{I,ini}$ and $\gamma_{I,ini}$ using the relation $\underline{v_I} = \underline{v_R} + \underline{\omega_E} \times \underline{r_E}$; the two values $v_{R,ini}$ and $\gamma_{R,ini}$ coincide with v_{E1} and γ_{E1} , used as initial conditions for reentry.
- Propagate numerically the reentry trajectory up to touchdown (altitude = 0) using equation set (B1); portray the time histories of $h(t)$, $\gamma_R(t)$, $v_R(t)$, and the reentry trajectory.

(d) Report the value of the time of flight, and the values of $a_d^{(\max)}$, $q_s^{(\max)}$, $p_d^{(\max)}$ and the times at which they occur.

For a_d , use
$$a_d = -\dot{v}_I = -\frac{d}{dt} \sqrt{v_R^2 + \omega_E^2 R^2 + 2\omega_E R v_R c_{\delta_R}}$$

(e) Portray $\delta_R(h)$, $v_R(h)$, $q_s(h)$, $a_d(h)$, $p_d(h)$

for the numerically integrated trajectory found at previous points and (in the same figures) the respective plots obtained with the use of the Allen-Eggers analytical solution.

For the evaluation of q_s , use the following constant

$$\frac{\kappa}{v_0^3 \sqrt{\rho_0}} = 1.762 \cdot 10^{-4} \text{ Kg}^{1/2} \text{ m}^{-1}$$